

LLM Post-training Data Selection

A controlled study of budget-equivalent selection and candidate-utility reliability

RESEARCH QUESTION

Under matched sample count, response-supervision tokens, and data composition, does targeted instruction selection outperform matched random selection?

Controlled design

2

selection methods

4

list realizations per method

3

training seeds

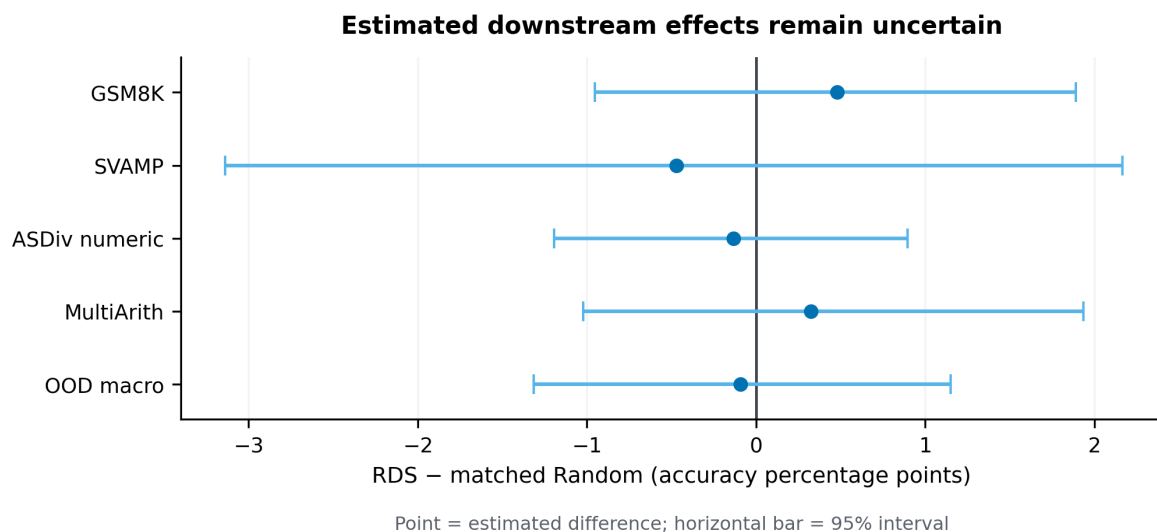
24

audited cells

MATCHED FACTORS

- 500 examples, response-supervision exposure, source x answer-length composition
- same base model, LoRA recipe, optimizer protocol, parser, and evaluation

Main result



BOUNDED INTERPRETATION

GSM8K: +0.480 pp

95% interval [-0.954, +1.889] pp

OOD macro: -0.094 pp

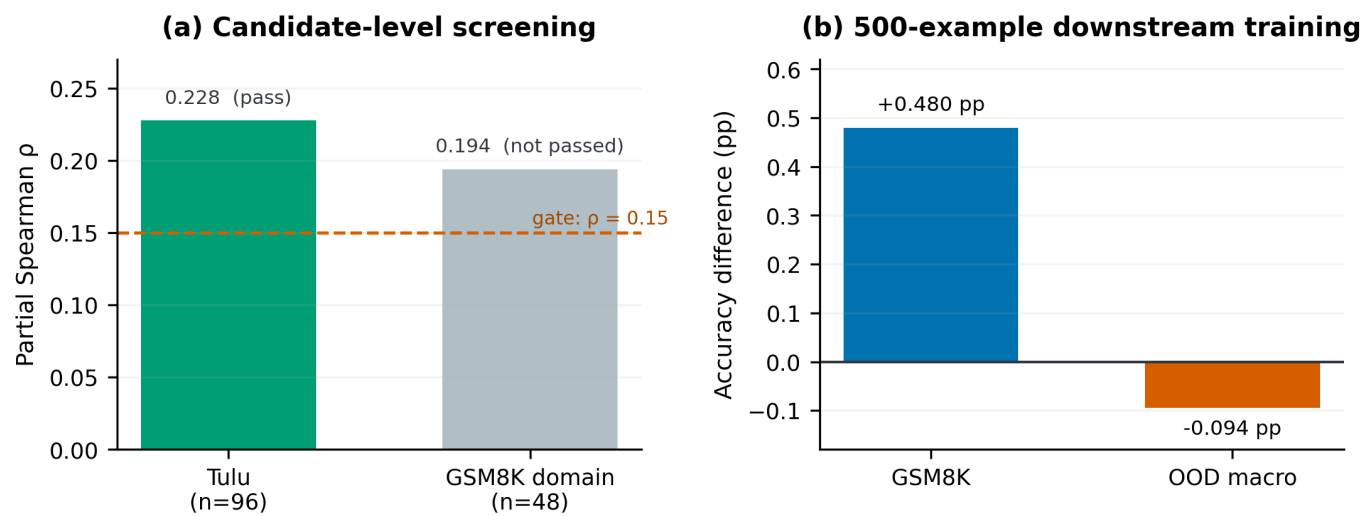
95% interval [-1.316, +1.149] pp

- No reliable downstream advantage was observed for the frozen RDS policy.
- The intervals still include relevant positive and negative effects.
- The study therefore establishes neither ineffectiveness nor equivalence.

From candidate signal to a falsifiable next study

Why a local screening result did not become a stable set-level benefit

Candidate-level signal did not yield a reliable set-level gain

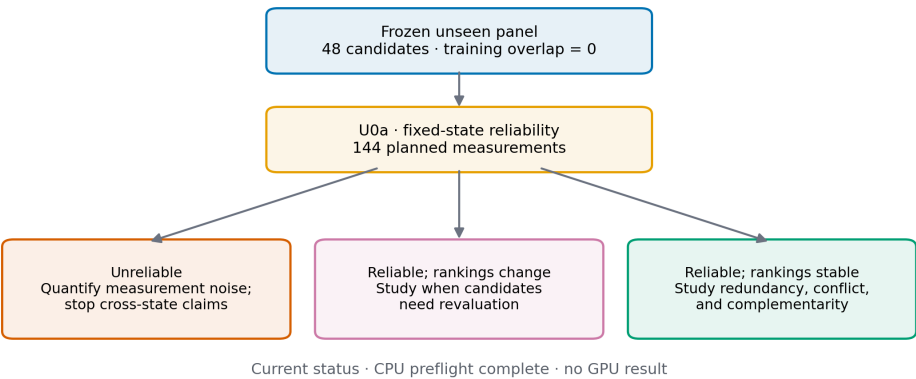


OBSERVED GAP

Tulu96 passed its original screen (partial Spearman 0.228, $p = 0.073$), but the 500-example downstream study did not show a stable gain.

Frozen next stage

State Dependence v3: measure reliability before mechanism



STATE DEPENDENCE v3

- 48 frozen candidates
- training overlap = 0
- U0a = 144 probes
- CPU preflight complete
- no GPU result

STOP/GO ORDER IS FROZEN

Four-week research module

